

PROJECT DESIGN PHASE

PROPOSED SOLUTION

Date	3rd October 2026
Team ID	6ab8c0aa8c7c66e70673078c
Project Name	Script-Controlled ACL – Restrict Record Access Based on Field Value
Maximum Marks	

4.2 PROPOSED SOLUTION

The proposed solution is a ServiceNow-based record-level access-control mechanism using custom roles and ACLs. The solution is implemented using the Institution Details (u_institution_details) table and four custom roles.

Main Components

1. Test User

A test user named EEE User is created to verify the behavior of the configured access controls.

2. Custom Roles

Four custom roles are created:

- bb1
- bb2
- bb3
- bb4

3. Institution Details Table

The custom table is configured as:

Label: Institution Details

Name: u_institution_details

The table contains:

- Student Roll Number – Auto Number
- Student Name – Reference to User
- Faculty Name – Reference to User
- Branch – Choice
- Email – String
- Phone Number – String
- Description – Multi String

The Branch field contains:

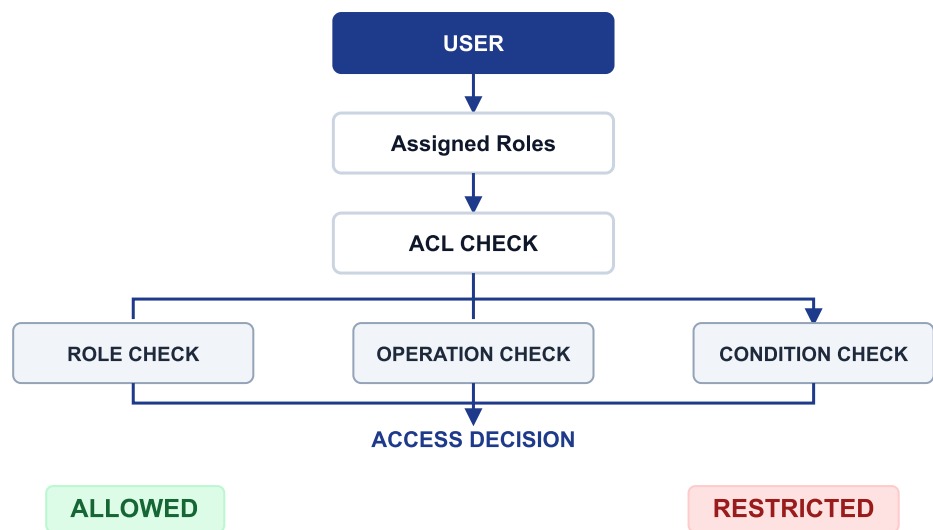
- ECE
- EEE
- CSE

4. ACL Configuration

Four separate Record ACLs are configured.

ACL	Operation	Required Role	Condition
READ ACL	Read	bb1	Branch = EEE
CREATE ACL	Create	bb2	No additional data condition
WRITE ACL	Write	bb3	No additional data condition
DELETE ACL	Delete	bb4	No additional data condition

Proposed Access Model



Progressive Permission Model

The permissions are implemented progressively:

Role Combination	Access Provided
bb1	Read permitted EEE records
bb1 + bb2	Read + Create
bb1 + bb2 + bb3	Read + Create + Write
bb1 + bb2 + bb3 + bb4	Read + Create + Write + Delete

Proposed Solution Flow

